

100 Essential Statistics Terms Used in Machine Learning

Example:

ID	Study Hrs/Week	Sleep Hrs/Night	Score	Pass/Fail
001	12	7	88	Pass
002	5	5	45	Fail
003	15	8	95	Pass
004	3	6	40	Fail
005	8	7	75	Pass

▮ Descriptive Stats

1. **Mean:** The average of all exam scores. $\rightarrow (88+45+95+40+75)/5 = 68.6$
2. **Median:** Middle value when scores are sorted. $\rightarrow 40, 45, 75, 88, 95 \rightarrow 75$
3. **Mode:** Most frequent score. $\rightarrow$ All unique $\rightarrow$ **None**
4. **Standard Deviation:** Spread from the mean. $\rightarrow \sim 21.7$
5. **Variance:** Std Dev² $\rightarrow 471$
6. **Range:** Max - Min $\rightarrow 95 - 40 = 55$
7. **Min/Max:** Min = **40**, Max = **95**
8. **Percentile:** 88 in 80th percentile = 80% students scored < 88
9. **Quartiles:** Q1 = 45, Q3 = 88
10. **IQR:** Q3 - Q1 = **43**
11. **Outlier:** A score like **15** would be one (not present)
12. **Z-Score:** $(95 - 68.6)/21.7 = 1.2$
13. **Skewness:** If most students score high, it's left-skewed
14. **Kurtosis:** High if scores are tightly grouped
15. **Distribution:** Overall shape $\rightarrow \sim$ Normal

▮ Probability Distributions

- 16. **Normal Distribution:** Most scores near mean → Bell curve
- 17. **Uniform Distribution:** All scores equally likely → Not seen here
- 18. **Binomial Distribution:** For Pass/Fail outcomes → Yes
- 19. **Bernoulli Distribution:** One trial of Pass/Fail → Each row
- 20. **Poisson Distribution:** Rare events like 100s/day → Not shown

▮ Probability Concepts

- 21. **Probability:** $P(\text{Pass}) = 3/5 = \mathbf{0.6}$
- 22. **Conditional Probability:** $P(\text{Pass} \mid \text{Study} > 10) = 2/2 = \mathbf{1.0}$
- 23. **Joint Probability:** $P(\text{Study} > 10 \cap \text{Pass}) = 2/5 = \mathbf{0.4}$
- 24. **Marginal Probability:** $P(\text{Pass})$ regardless of other variables = **0.6**
- 25. **Bayes' Theorem:** Update $P(\text{Pass})$ given Study Hours
- 26. **Event:** One outcome, e.g., Score > 90
- 27. **Sample Space:** All Pass/Fail outcomes = {Pass, Fail}
- 28. **Random Variable:** Score value per student
- 29. **Independent Events:** One student's pass ≠ affect others
- 30. **Dependent Events:** Sleep hours may affect Score
- 31. **Mutually Exclusive:** A student can't both Pass and Fail

▮ Hypothesis Testing

- 32. **Hypothesis:** "More study improves score"
- 33. **Null Hypothesis (H_0):** Study hrs have no effect
- 34. **Alternative Hypothesis (H_1):** Study hrs do affect score
- 35. **p-value:** Probability result occurred under H_0 . If $p = 0.03$, reject H_0
- 36. **Alpha:** Significance level (usually 0.05)
- 37. **Confidence Interval (CI):** Score CI = [65, 72]
- 38. **T-test:** Compare mean scores of high vs low study hrs
- 39. **Chi-Square Test:** Study Pass vs Fail across categories
- 40. **ANOVA:** Compare means for 3+ sleep groups

▮ Correlation & Regression

- 41. **Correlation:** Study hrs ↑ → Score ↑ → $r = \mathbf{0.7}$
- 42. **Pearson Correlation:** Measures linear relationships
- 43. **Spearman Correlation:** Based on rank if data is non-linear

- 44. **Covariance:** If study $\uparrow$ and score $\uparrow$, it's positive
- 45. **Regression:** Predict score using study & sleep hrs
- 46. **Linear Regression:** $\text{score} = a + b_1(\text{study}) + b_2(\text{sleep})$
- 47. **Logistic Regression:** Predict Pass/Fail
- 48. **Residual:** Actual score - Predicted score

▮ ML Model Behavior

- 49. **Overfitting:** Model learns noise, e.g., memorizing scores
- 50. **Underfitting:** Model too simple to capture trends
- 51. **Bias:** Error due to wrong assumptions
- 52. **Variance:** Sensitivity to training set changes
- 53. **Bias-Variance Tradeoff:** Balance both for best model

✓ Model Evaluation

- 54. **Accuracy:** $(TP+TN)/\text{Total} = (3+1)/5 = \mathbf{0.8}$
- 55. **Precision:** $TP/(TP+FP) = 3/(3+1) = \mathbf{0.75}$
- 56. **Recall:** $TP/(TP+FN) = 3/(3+1) = \mathbf{0.75}$
- 57. **F1 Score:** Harmonic mean of precision & recall $\rightarrow \sim \mathbf{0.75}$
- 58. **Confusion Matrix:**

	Pred: Pass	Pred: Fail
Actual Pass	TP = 3	FN = 1
Actual Fail	FP = 1	TN = 0
- 59. **TP:** Predicted Pass and was Pass $\rightarrow$ 3 students
- 60. **FP:** Predicted Pass but was Fail $\rightarrow$ 1 student
- 61. **TN:** Predicted Fail and was Fail $\rightarrow$ 0
- 62. **FN:** Predicted Fail but was Pass $\rightarrow$ 1 student

▮ Sampling & Inference

- 63. **Sampling:** Pick 100 students from 1,000
- 64. **Random Sampling:** Equal chance for all
- 65. **Stratified Sampling:** Balanced by gender or grade
- 66. **Bootstrap Sampling:** With replacement $\rightarrow$ resample students
- 67. **Population:** All 1,000 students
- 68. **Sample:** Subset like 5 shown
- 69. **Central Limit Theorem:** Means from samples $\approx$ normal
- 70. **Law of Large Numbers:** Bigger sample $\rightarrow$ closer to true mean

- 71. **Parameter:** True average score in population
- 72. **Statistic:** Sample mean → 68.6
- 73. **Estimator:** Method for estimating mean
- 74. **Likelihood:** How likely observed scores are under model
- 75. **MLE:** Best parameters that maximize likelihood of this data
- 76. **Prior Probability:** $P(\text{Pass})$ before seeing study hrs
- 77. **Posterior Probability:** $P(\text{Pass}|\text{study hrs})$

▮ Information Theory

- 78. **Entropy:** Uncertainty in pass/fail → High if split is 50/50
- 79. **Cross-Entropy:** Measure classification error
- 80. **Information Gain:** Reduction in entropy after split
- 81. **Gini Index:** Impurity used in decision trees

▮ Data Preprocessing

- 82. **Feature Scaling:** Normalize study hours
- 83. **Normalization:** Scale 0–1 range
- 84. **Standardization:** Convert to Z-scores
- 85. **Outlier Detection:** 150 study hours would be one
- 86. **Missing Data:** If sleep data not recorded
- 87. **Imputation:** Fill missing with average sleep hrs = 6.6

▮ Visuals & Distributions

- 88. **PDF:** Smooth curve of score probabilities
- 89. **CDF:** Accumulated % of students by score
- 90. **Histogram:** Bars of score frequency
- 91. **Box Plot:** Visual of Q1, Median, Q3 & outliers
- 92. **QQ Plot:** Compares score distribution to normal

▮ Classification Metrics

- 93. **ROC Curve:** TPR vs FPR curve
- 94. **AUC:** Area under ROC curve → 0.85 is good
- 95. **Multicollinearity:** If study & sleep are highly correlated

📊 Time Series Stats

- 96. **Autocorrelation:** Score trend over months
- 97. **Time Series:** Track weekly scores per student
- 98. **Stationarity:** Mean/variance stay same over time
- 99. **Smoothing:** Use moving average to show trends

✓ Final One:

- 100. **Simplicity Wins:** Always explain stats with clear examples like this 📊

